

بیوشیمی نظری گروه الف

نیمسال دوم سال تحصیلی ۱۴۰۱-۱۴۰۲

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زمان آزمون: ۳۰ دقیقه تاریخ آزمون: ۱۴۰۲/۴/۲۵

- مشخصات دانشجویی خود را روی صفحه اول پرسشنامه بنویسید.
- لطفاً سوالات پرسشنامه را به دقت مطالعه، سپس گزینه صحیح را در پاسخنامه با مداد مشکی وارد نمایید.
- از نوشتن هرگونه توضیح اضافی در پاسخنامه و پرسشنامه جداً خودداری نمایید.
- ۳۹ سوال تستی در ۴ صفحه تهیه شده است. لطفاً تعداد صفحات را کنترل نمایید.

1. Regarding **hexokinase and glucokinase**, all of the following options are correct, **EXCEPT**:

- a) The V_m of hexokinase is greater than the V_m of glucokinase.
- b) The K_m of glucokinase is greater than the K_m of hexokinase.
- c) An increase in fructose-6-phosphate leads to inhibition of the glucokinase enzyme.
- d) An increase in glucose-6-phosphate causes inhibition of hexokinase.

2. Regarding the **pyruvate dehydrogenase enzyme complex**, all of the following are correct, **EXCEPT**:

- a) In the first enzymatic component of the complex, CO_2 is released.
- b) Its final reducing coenzyme is FADH_2 .
- c) An increase in pyruvate leads to inhibition of the enzyme pyruvate dehydrogenase kinase.
- d) With an increase in the acetyl-CoA/CoA ratio, it is allosterically inactivated.

3. A person who had malaria developed hemolytic anemia after receiving the drug **primaquine**. A deficiency of which of the following could have led to hemolysis?

- a) Ribose-5-phosphate
- b) Glucose-6-phosphate
- c) NADH
- d) Reduced glutathione (GSH)

4. The transport of **monosaccharides** from the intestinal mucosal cells into the portal bloodstream occurs through which transporter?

- a) GLUT-1
- b) GLUT-2
- c) GLUT-3
- d) GLUT-4

5. In the Krebs cycle, from **alpha-ketoglutarate to succinate**, how many ATP equivalents are ultimately produced?

- a) 3
- b) 4
- c) 5
- d) 6

6. An increase in which of the following leads to activation of **pyruvate kinase**?

- a) Fructose-1,6-bisphosphate
- b) Phosphorylation
- c) ATP
- d) Alanine

7. The conversion of which of the following compounds (in a single reaction) is seen **only in gluconeogenesis**?

- a) Oxaloacetate and phosphoenolpyruvate
- b) Phosphoenolpyruvate and pyruvate
- c) Glucose-6-phosphate and fructose-6-phosphate
- d) 1,3-bisphosphoglycerate and 3-phosphoglycerate

8. Regarding **fructose and galactose metabolism**, which of the following statements is correct?

- a) Glucose and galactose are interconverted by the enzyme 4-epimerase.
- b) Aldolase B deficiency leads to essential fructosuria.
- c) The complications of classic galactosemia are more severe than those of galactokinase deficiency.
- d) Hereditary fructose intolerance arises from fructokinase deficiency.

9. In which of the following **glycogen storage diseases** is there a deficiency of **muscle glycogen phosphorylase**?

- a) Forbes or Cori (Type III)
- b) Pompe (Type II)
- c) Hers (Type VI)
- d) McArdle syndrome (Type V)

10. The formation of which of the **urea cycle intermediates** is energy-requiring (endergonic)?

- a) Argininosuccinate
- b) Citrulline
- c) Ornithine
- d) Arginine

11. All of the following enzymes require **tetrahydrobiopterin (BH4)** for their function, **EXCEPT**:

- a) Tryptophan hydroxylase
- b) Phenylalanine hydroxylase
- c) Dopamine hydroxylase
- d) Tyrosine hydroxylase

12. A deficiency of the enzyme **homogentisate dioxidase (homogentisic acid oxidase)** results in which disease?

- a) Maple syrup urine disease (MSUD)
- b) Homocystinuria
- c) Phenylketonuria
- d) Alkaptonuria

13. In all of the following, the three amino acids can be converted, respectively, to **pyruvate, succinyl-CoA, and alpha-ketoglutarate, EXCEPT:**

- a) Alanine, tyrosine, arginine
- b) Cysteine, isoleucine, proline
- c) Serine, valine, histidine
- d) Glycine, threonine, glutamine

14. For the conversion of **alpha-ketoglutarate to glutamate** by the enzyme **glutamate dehydrogenase**, which coenzyme is required?

- a) PLP
- b) FAD
- c) NADPH
- d) NADH

15. Regarding the synthesis of **asparagine and glutamine**, which of the following statements is correct?

- a) In the synthesis of asparagine, ammonia is used as the amine source.
- b) The synthesis of asparagine and glutamine requires ATP as the energy source.
- c) In the synthesis of glutamine, asparagine is used as the amine source.
- d) The synthesis of glutamine requires the breaking of two high-energy bonds.

16. The measurement of which of the following enzymes has **no clinical value in muscle diseases**?

- a) Aspartate aminotransferase (AST)
- b) Lactate dehydrogenase (LDH)
- c) Alkaline phosphatase (ALP)
- d) Aldolase (ALS)

17. Which of the following enzymes is present in **large amounts in red blood cells**, so a hemolyzed sample should not be used?

- a) Lactate dehydrogenase (LDH)
- b) Creatine phosphokinase (CPK)
- c) Alkaline phosphatase (ALP)
- d) Alanine aminotransferase (ALT)

18. To distinguish **pulmonary infarction from myocardial (cardiac) infarction**, measurement of which pair of enzymes is useful?

- a) Alanine aminotransferase (ALT) and alkaline phosphatase (ALP)
- b) Creatine phosphokinase (CPK) and gamma-glutamyl transferase (GGT)
- c) Lactate dehydrogenase (LDH) and alkaline phosphatase (ALP)
- d) Aspartate aminotransferase (AST) and lactate dehydrogenase (LDH)

19. All of the following statements are correct, **EXCEPT**:

- a) The hormone secretin induces the secretion of bicarbonate from the gallbladder.
- b) Lingual lipase separates short-chain fatty acids from triglyceride.
- c) Short-chain fatty acids are transported in the blood bound to albumin.
- d) Lipoprotein lipase plays a role in chylomicron metabolism.

20. A 45-year-old man with a history of one heart attack in the past six months presents to a physician with an LDL level of 210 mg/dL and triglycerides of 140 mg/dL. For a definitive diagnosis of the disease by identifying possible mutations, **which gene's sequencing would you recommend?**

- a) LDL Receptor
- b) Apo E
- c) LPL
- d) Apo B48

21. All of the following factors play a role in **chylomicron metabolism**, **EXCEPT**:

- a) LPL
- b) CETP
- c) HDL-C
- d) Apo CII

22. Increased triglyceride together with decreased **HDL-C** is seen in all of the following, **EXCEPT**:

- a) Pregnancy
- b) Stress
- c) Type 2 diabetes
- d) Hypothyroidism

23. Which option is correct in the **fed (satiety) state**?

- a) Decreased activity of fatty acid synthase
- b) Decreased activity of acetyl-CoA carboxylase
- c) Decreased activity of carnitine palmitoyltransferase I (CPT-I)
- d) Decreased activity of HMG-CoA reductase

24. Which one causes **activation of the enzyme HMG-CoA reductase**?

- a) cAMP
- b) Atorvastatin
- c) Insulin
- d) Protein Kinase

25. From the **β -oxidation of myristic acid** in the mitochondria, all of the following compounds are produced, **EXCEPT**:

- a) Malonyl CoA
- b) Acetyl CoA
- c) FADH_2
- d) NADH

26. Which compound plays a role in the transport of **acetyl-CoA from the mitochondria to the cytosol**?

- a) Malate
- b) Carnitine
- c) Pyruvate
- d) Citrate

27. The synthesis of **palmitic acid** requires all of the following, **EXCEPT**:

- a) Carnitine
- b) Acetyl CoA
- c) ATP
- d) $\text{NADPH} + \text{H}^+$

28. In which of the following conditions is **urinary urobilinogen increased**?

- a) Dubin–Johnson syndrome
- b) Biliary duct obstruction
- c) Hepatitic jaundice
- d) Hemolytic anemia

29. A hereditary defect or absence of which enzyme causes **immunodeficiency in newborns**?

- a) Thymidylate synthetase
- b) Adenosine deaminase
- c) Xanthine oxidase
- d) Ribonucleotide reductase

30. All of the following compounds participate in the **biosynthesis of pyrimidine nucleotides**, **EXCEPT**:

- a) Aspartate
- b) Phosphoribosyl pyrophosphate
- c) Carbamoyl phosphate
- d) Glycine

31. **5-fluorouracil** is an inhibitor of which of the following enzymes?

- a) Ribonucleotide reductase
- b) Dihydrofolate reductase
- c) Thymidine synthase (thymidylate synthase)
- d) Xanthine oxidase

32. Porphyrria cutanea tarda is caused by a defect in which of the following enzymes?

- a) ALA synthase
- b) Ferrochelatase
- c) Uroporphyrinogen decarboxylase
- d) Coproporphyrinogen oxidase

33. Lesch–Nyhan syndrome, which is an X-linked recessive disorder, is caused by a deficiency of which enzyme?

- a) Hypoxanthine-guanine phosphoribosyl transferase (HGPRT)
- b) Adenosine deaminase
- c) Xanthine oxidase
- d) Adenine phosphoribosyl transferase

34. Which of the respiratory chain complexes does **NOT** act as a **proton pump**?

- a) Succinate Q reductase
- b) NADH dehydrogenase
- c) Cytochrome C oxidase
- d) Cyt C reductase

35. The **ATP/ADP transporter** in the inner mitochondrial membrane is inhibited by

- a) Oligomycin
- b) CN^-
- c) Atractyloside
- d) Barbiturates

36. Which part of the respiratory chain has the **lowest oxidation potential**?

- a) FMN
- b) Coenzyme Q
- c) Cyt b
- d) Cyt c

37. All of the following occur in the **fasting state**, **EXCEPT**:

- a) Epinephrine stimulates glycogenolysis in the liver.
- b) Glucagon stimulates gluconeogenesis in the liver.
- c) Glucagon stimulates glycogenesis in the liver.
- d) Insulin is released at basal levels from pancreatic cells.

38. All of the following statements regarding the **use or production of ketone bodies by body tissues** are correct, **EXCEPT**:

- a) Skeletal muscle — ketone body production
- b) Liver — production
- c) Cardiac muscle — use
- d) Kidney — use

39. Which of the following statements about **glucose metabolism in a Type 1 diabetic** person is correct?

- a) Glucose uptake in intestinal epithelial cells is impaired.
- b) Glucose uptake by skeletal muscle cells occurs easily.
- c) Glucose uptake by adipose tissue cells occurs easily.
- d) It has no effect on glucose reabsorption from the renal tubules.

